Economic Growth and Agricultural Development:

How GDP Performance Correlates with Food Security Across Sub-Saharan Africa

AnalystLab Africa Data Analytics Capstone Project
Week 8 | Batch B (June–August 2026)
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Date: July 25, 2026

Executive Summary

This analysis reveals a strong positive correlation between GDP growth and improved development outcomes across Sub-Saharan Africa over the 1960–2023 period. Through comprehensive examination of World Bank development indicators across 50 SSA countries, we found that economic growth has contributed to gains in life expectancy (currently 64.80 years) and other development metrics. However, despite sustained GDP growth since 1960, food insecurity remains alarmingly high, affecting 58.32% of the population on average. The research demonstrates that while GDP growth is foundational to development progress, targeted agricultural investment and food systems strengthening are essential complementary strategies to achieve meaningful food security gains.

1. Objective & Problem Statement

This capstone project investigates the relationship between economic performance—measured through GDP and related indicators—and food security outcomes in Sub-Saharan Africa. The analysis addresses a critical development question: To what extent does macroeconomic growth translate into improvements in food security and agricultural productivity? By examining 63 years of development data across 50 SSA nations, this project aims to identify patterns and correlations that inform policy decisions for development organizations such as the FAO and WFP. The research seeks to understand which economic pathways most effectively support food security and to identify countries that have successfully leveraged economic growth for development gains.

2. Dataset Description

Data Source

World Bank World Development Indicators (WDI) – The world's most comprehensive collection of global development statistics, containing over 1,400 indicators covering economic, social, environmental, and institutional dimensions of development.

Dataset Scope

Geographic Coverage: 50 Sub-Saharan African countries (Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Comoros, Congo, Democratic Republic of Congo, Côte d'Ivoire, Djibouti, Equatorial Guinea, Eritrea, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Rwanda, São Tomé and Príncipe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, Sudan, Swaziland, Tanzania, Togo, Uganda, Zambia, Zimbabwe)

Time Period: 1960–2023 (63 years of longitudinal data)

Total Records: 15,225 data points

Indicators Analyzed: 7 core development metrics

Key Indicators Analyzed

1. GDP (current US$) – Total gross domestic product measured in nominal US dollars, reflecting absolute economic size2. GDP per capita (current US$) – Economic output per person, indicating prosperity levels3. Agriculture, forestry, and fishing (% of GDP) – Agricultural sector's contribution to total economy4. Food production index (2014-2016=100) – Agricultural productivity trends5. Prevalence of moderate or severe food insecurity (%) – Proportion of population experiencing food insecurity6. Poverty headcount ratio at $1.90 a

day (%) – Extreme poverty incidence7. Life expectancy at birth, total (years) – Overall development and health indicator

3. Data Cleaning Process

The World Development Indicators dataset, while comprehensive, required systematic cleaning to ensure analytical quality:

Step 1 – Indicator Selection: From 1,400+ available indicators, 7 core metrics were selected based on relevance to the research question and data availability.

Step 2 – Geographic Filtering: Data was filtered to include only the 50 Sub-Saharan African countries, excluding North African nations and non-African countries.

Step 3 – Missing Value Handling: Records with missing values (NaN) were identified and removed. Early years (1960–1980) had significant data gaps, particularly for food security indicators which are only available from 2014 onwards.

Step 4 – Format Transformation: Data was converted from wide format (years as columns) to long format (Year, Country, Indicator, Value as rows) to enable time-series analysis and Power BI visualization.

Step 5 – Data Validation: Numeric values were verified for consistency, outliers were examined, and data types were standardized.

Step 6 – Final Dataset: The cleaned dataset resulted in 15,225 valid records spanning 1960–2023, ready for analysis.

4. Analysis Methodology

This analysis employed multiple analytical techniques to examine economic and food security trends:

Descriptive Statistics: Key Performance Indicators (KPIs) were calculated for 2023 (the latest available year with complete data across all indicators). These KPIs provide a snapshot of current development status across the 50-country sample.

Trend Analysis: Historical trends were examined from 1960–2023 to identify periods of economic growth, stagnation, or decline. This longitudinal perspective reveals how development outcomes have evolved over six decades.

Correlation Analysis: Scatter plot visualization was employed to examine the relationship between GDP and other development outcomes. A positive correlation would indicate that wealthier countries experience better development indicators.

Comparative Analysis: Countries were ranked by agricultural contribution to GDP, revealing the relative importance of agriculture across SSA economies.

Visualization: Interactive Power BI dashboard was developed to allow stakeholders to explore data by country, year, and indicator, enabling deeper investigation of patterns.

5. Key Findings

Finding 1: Life Expectancy Gains

Average life expectancy across the 50 SSA countries stands at 64.80 years (2023 data). This represents substantial improvement from the 1960s when life expectancy was significantly lower. This metric indicates that health and development outcomes have improved alongside economic growth.

Finding 2: Persistent Food Insecurity

Despite economic growth, 58.32% of the SSA population experiences moderate or severe food insecurity (2023). This alarming figure reveals a critical gap: macroeconomic growth has not automatically translated into food security for the majority of the region's inhabitants.

Finding 3: Agriculture's Varied Contribution

Agriculture contributes an average of 19.25% to SSA GDP. However, this varies dramatically by country: Niger has the highest agricultural contribution to GDP, reflecting its agriculture-dependent economy, while Djibouti has the lowest, indicating a more service and trade-oriented economy.

Finding 4: Substantial Regional GDP Growth

Total SSA GDP reached $47.02 billion (2023). Analysis of the 1960–2023 period reveals consistent economic growth, particularly accelerating from the 1990s onwards. This growth reflects expanding economies, increased trade, and development investments across the region.

Finding 5: Positive Correlation Between GDP and Development Outcomes

The scatter plot analysis reveals a strong positive correlation between GDP and development outcomes. Countries with higher GDP levels generally demonstrate better outcomes across multiple indicators including life expectancy and other development metrics. This relationship validates the economic growth hypothesis.

6. Insights & Analysis

Insight 1: Economic Growth is Necessary but Not Sufficient

The data reveals a paradox: despite 63 years of economic growth and a current GDP of $47.02 billion, nearly 60% of SSA's population still faces food insecurity. This indicates that GDP growth alone—without targeted interventions in agriculture and food systems—is insufficient to achieve food security. Economic growth provides the financial resources and foundation for development, but these resources must be strategically deployed to address specific development challenges.

Insight 2: Agriculture Remains Critical Despite Declining Share

While agriculture's share of GDP averages 19.25% and is declining as economies diversify, it remains critical for food security. Countries like Niger, where agriculture represents a substantial portion of GDP, are vulnerable to agricultural shocks. Conversely, countries like Djibouti with service-based economies may still face food insecurity if they lack sufficient agricultural productivity and import capacity.

Insight 3: Development Progress is Non-Linear

The GDP trend line (1960–2023) shows acceleration particularly from 1990 onwards, reflecting post-Cold War economic liberalization, increased trade, and development investments. However, the trend is not uniformly positive across all countries—some experience declines during crisis periods. This heterogeneity suggests that country-specific factors (governance, natural resources, regional conflicts) significantly influence outcomes.

7. Recommendations

Recommendation 1: For International Development Organizations (FAO, WFP)

Prioritize investment in smallholder agriculture and food systems in countries where food insecurity excee ds 50%. Target countries with lower agricultural productivity (indicated by food production indices) to unlock productivity gains. Use the GDP-to-food-security correlation to identify countries where economic growth could translate to food security improvements with targeted support. Consider establishing multi-country programs that combine economic growth support with specific food security interventions.

Recommendation 2: For SSA National Governments

Develop integrated policies that explicitly link macroeconomic growth targets to food security outcomes. Rather than treating GDP growth and food security as separate policy domains, create comprehensive development strategies that ensure growth benefits reach agricultural sectors and food-insecure populations. Invest in agricultural extension services, market infrastructure, and climate-resilient farming practices. Use subsidy and tax mechanisms to support smallholder farmers while pursuing broader economic growth.

Recommendation 3: For Data Monitoring and Impact Evaluation

Establish regular tracking of the GDP-to-food-security correlation at country and sub-regional levels. This monitoring framework should identify intervention points and measure policy effectiveness. Create data dashboards (similar to the one developed in this capstone) that allow policymakers to monitor progress toward Sustainable Development Goals, particularly SDG 1 (No Poverty) and SDG 2 (Zero Hunger).

Recommendation 4: For Future Research and Analysis

Investigate lag effects between GDP growth and food security improvements. Some development outcomes take years to materialize. Additionally, conduct sector-specific analysis to identify which economic sectors (trade, technology, infrastructure, manufacturing) most effectively contribute to food security. Include qualitative research on countries that have successfully translated economic growth into food security to identify best practices.

8. Conclusion

This capstone project demonstrates that Sub-Saharan Africa has experienced significant economic growth over the past six decades, with GDP expanding from minimal levels in 1960 to $47.02 billion by 2023. This growth has supported improvements in life expectancy (currently 64.80 years) and other development indicators. The strong positive correlation between GDP and development outcomes validates the fundamental economic development hypothesis: growth creates the foundation for progress.

However, the persistent high prevalence of food insecurity (58.32%) reveals that macroeconomic growth alone is insufficient to achieve development goals. Despite six decades of economic expansion, the majority of SSA's population still lacks reliable access to adequate food. This gap between economic growth and food security outcomes highlights the critical need for targeted interventions that specifically address agricultural productivity, food systems resilience, and social protection.

The varied contribution of agriculture across countries (from Niger's high dependence to Djibouti's service-based economy) indicates that effective strategies must be country-specific, accounting for each nation's economic structure, natural resources, and institutional capacity.

For development organizations like the FAO and WFP, this analysis provides a clear imperative: leverage the region's economic growth momentum through strategic investments in agricultural development and food systems. The data-driven insights from this capstone demonstrate that evidence-based policymaking—grounded in rigorous analysis of the relationship between economic and social outcomes—is essential for achieving the Sustainable Development Goals and building more prosperous, food-secure futures for Sub-Saharan Africa's populations.

By pursuing integrated strategies that combine macroeconomic growth with targeted food security investments, SSA nations can translate economic progress into meaningful improvements in human wellbeing and food security for all.

Appendix: Dashboard & Methodology

An interactive Power BI dashboard was developed to visualize the analysis findings. The dashboard includes:

• KPI Cards: Displaying 2023 values for Life Expectancy (64.80 years), Food Insecurity (58.32%), Agriculture % of GDP (19.25%), and Total GDP ($47.02bn)• GDP Trend Line Chart: Showing historical GDP trends from 1960–2023 across all 50 countries• Scatter Plot: Illustrating the correlation between GDP and development outcomes• Agricultural Contribution Bar Chart: Ranking countries by agriculture's contribution to GDP, with Niger highest and Djibouti lowest• Interactive Slicers: Allowing exploration by country, year range, and indicatorThis dashboard serves as a tool for stakeholders to explore data independently and drill down into country-specific or regional patterns.

All analysis was conducted using Python (Pandas for data manipulation) and Microsoft Power BI for visualization. The analysis adheres to best practices in data cleaning, exploratory data analysis, and descriptive statistics.